

Krishna Satya Koppineedi

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SUMMARY

AI/ML Engineer with 4+ years of experience delivering scalable, real-time AI, generative AI, and agentic AI solutions in financial services and retail applications. Expert in Python, PyTorch (Lightning), Hugging Face Transformers, RAG, LangChain, Graph ML, XGBoost, and vector databases (FAISS, Weaviate, Milvus). Skilled in predictive modeling, anomaly detection, prompt engineering, GPU-optimized inference, streaming ML, and MLOps, building production-ready pipelines that improve trading strategies, fraud detection, and customer personalization.

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch, PyTorch Lightning, TensorFlow/Keras, FastAPI, REST APIs, Prompt Engineering, Scala
AI/ML & Data Engineering: LSTM, GRU, 1D-CNN, CNN, Vision Transformers, Transformer-based LLMs (BERT/GPT), Hugging Face, RAG, LangChain, LoRA/PEFT fine-tuning, embeddings, Graph ML (NetworkX, PyTorch Geometric), FAISS, Weaviate, ONNX, TensorRT, GPU optimization, Self-supervised Learning, Reinforcement Learning, Agentic AI, Generative AI orchestration
Cloud & MLOps: AWS (SageMaker, EC2, Lambda, S3), Azure ML, Docker, Kubernetes, MLflow, Apache Airflow, CI/CD pipelines, model drift detection, Canary deployments, real-time monitoring & auto-retraining, streaming ML pipelines (Kafka + Spark Structured Streaming)
Databases & Data Management: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, Memcached, SQL query optimization
Visualization & Monitoring: Power BI, Tableau, Advanced Excel, REST APIs, Explainable AI (SHAP, Captum), real-time alerting
Collaboration & Agile: Agile (Scrum/Kanban), JIRA, Confluence, cross-functional team collaboration

PROFESSIONAL EXPERIENCE

Goldman Sachs, Miami, Florida **Aug 2024 – Present**
AI/ML Engineer

- Orchestrated comprehensive market volatility forecasting pipelines in Python, integrating data from 10+ heterogeneous sources, performing preprocessing, feature engineering, and transformations, resulting in a 10% improvement in forecast accuracy and enabling more informed risk management decisions.
- Developed Transformer-based models (BERT/GPT) for textual market sentiment analysis on unstructured financial news and analyst reports, using RAG and LangChain for real-time context retrieval, which enabled 25% faster anomaly detection for trading signals and improved decision-making.
- Engineered time-series deep learning models (LSTMs, GRUs) for high-frequency trading datasets, implementing advanced lag-based features and multi-horizon forecasting, reducing RMSE by 12% and optimizing trade execution strategies under real-time constraints.
- Managed MLOps frameworks (MLflow, Airflow) for end-to-end model training, versioning, automated retraining, and deployment, maintaining 99% reliability in production environments with automated alerts for anomalies.
- Optimized low-latency GPU inference pipelines using ONNX and TensorRT, supporting sub-second predictions for market risk scoring and trade execution; integrated agentic AI to autonomously adjust models to market regime shifts and volatility changes.
- Designed anomaly detection frameworks combining ensemble tree-based methods (XGBoost, LightGBM) with deep neural networks, continuously monitored via real-time dashboards, reducing false positives for trading alerts by 20% and improving risk management accuracy.
- Deployed large-scale streaming pipelines using PySpark and Dask, serving ML models via FastAPI for real-time inference, increasing anomaly detection throughput by 25% while maintaining sub-second latency.
- Established LLM fine-tuning pipelines for domain-specific sentiment extraction, including evaluation using Rouge and BERTScore, improving trading signal reliability by 20% and enabling faster analyst decision-making.

LTIMindtree, India **Jan 2021 – Jul 2023**
Machine Learning Engineer

- Constructed a retail customer segmentation and product recommendation platform using LSTM, GRU, and transformer embeddings, analyzing purchase history, website interactions, and demographic features to identify high-value customer segments for personalized campaigns.
- Built an end-to-end personalized recommendation engine leveraging LangGraph embeddings and XGBoost scoring, dynamically generating product suggestions and improving customer engagement by 28%.
- Formed real-time inventory demand prediction models with PySpark streaming, predicting daily sales per product category, reducing stock-outs by 20% and optimizing inventory allocation across multiple retail stores.
- Delivered GPU-optimized model APIs with FastAPI and Docker, ensuring sub-second latency and 24/7 availability for personalized recommendation and demand prediction dashboards.
- Automated ML workflows using Apache Airflow and MLflow, including retraining triggers and version control, reducing model update cycles from weeks to under 24 hours while maintaining high data integrity.
- Built interactive monitoring dashboards using Power BI to track model performance, feature drift, and recommendation accuracy, achieving >99% reliability for production retail systems.
- Collaborated with cross-functional teams in Agile using JIRA, conducting code reviews, designing modular pipelines, and integrating ensemble models into customer-facing retail platforms, improving engagement and personalization metrics.

ACADEMIC PROJECTS

Generative AI-Powered Financial Insights using RAG Pipelines

- Implemented a retrieval-augmented generation (RAG) pipeline using Hugging Face Transformers and PyTorch to fetch relevant historical financial reports, enabling the LLM to generate context-aware summaries, improving analyst decision-making speed by 40%.
- Configured data preprocessing and feature extraction pipelines with NumPy and Pandas to structure unstructured news and analyst reports, optimizing input for generative AI models, resulting in a 25% increase in trend prediction accuracy.

Agentic AI-Driven Fraud Detection

- Programmed an agentic AI system that autonomously analyzes transactional graphs using NetworkX and PyTorch Geometric, learning account relationships and activity patterns, and generating fraud detection strategies in real time.
- Leveraged Pinecone vector DB to enable the agent to perform similarity searches on transaction embeddings, allowing it to proactively identify fraud patterns, reduce false positives by 30%, and trigger near real-time alerts.

Customer-Centric Banking Recommendation & Propensity Scoring System

- Architected customer embeddings with a large language model (LLM) and represented interactions as a LangGraph, capturing relationships in behaviors, transactions, demographics. Evaluated embeddings with cosine similarity, achieving a 22% improvement in personalized suggestion accuracy.
- Created an LLM-driven propensity scoring pipeline leveraging the LangGraph, combining transaction patterns and behavioral signals. Model performance was assessed using AUC-ROC and cross-sell uplift, achieving an 18% increase in predictive accuracy across key segments.

EDUCATION

Master of Science in Computer Science – (GPA 3.7/4.0)
University Of South Florida, Tampa, Florida